MIL-S-48464B(AR)
30 June 1981
SUPERSEDING
MIL-S-48464A(AR)
1 March 1976

# MILITARY SPECIFICATION

## SENSOR, CROSSWIND: 11733251

This specification is approved for use by the US Army Armament Research and Development Command (ARRADCOM) and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Sensor, Crosswind: 11733251 which is a unit of the M21 Ballistic Computer System, 11732979.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-F-13926 | Fire Control Materiel; General Specification Governing the Manufacture and Inspection of |
| MIL-I-45607 | Inspection Equipment, Acquisition, Maintenance and Disposition of |
| MIL-STD-45662 | Calibration System Requirements |

FSC 1220

**Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.**

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MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-810 | Environmental Test Methods |

DRAWINGS

ARRADCOM

| | |
|---|---|
| 11733251 | Sensor, Crosswind |

Inspection Equipment

| | |
|---|---|
| 11750233 | Test Set - Computer System FAT |
| 11822864 | Test Set - Crosswind Sensor |
| 11822890 | Shock and Vibration Fixture Assembly |
| 11822889 | Shock Fixture, Sensor Mast Mounting Plate |
| 11822891 | Cable, Wiring Diagram |
| 11733250 | Mast, Crosswind Sensor |
| 11821463 | Unit Fixture - Crosswind Sensor |
| 11821464 | Unit Fixture - Crosswind Sensor (-Z) |

Packaging Data Sheet

| | |
|---|---|
| P11733251 | Packaging of Sensor, Crosswind: 11733251 |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



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## 3. REQUIREMENTS

3.1 Fabrication. The crosswind sensor, herein referred to as either the assembly or CWS, shall be manufactured in accordance with Drawing 11733251 and drawings pertaining thereto and, when assembled, shall meet the requirements of this specification. (see 4.5.1)

3.1.1 Function. The assembly shall provide a voltage proportional to the component of wind normal to the line-of-sight.

3.1.2 General specifications. The following provisions of MIL-F-13926 apply: (See 4.5.1)

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.1.3 Ambient condtions. Standard ambient conditions shall be as follows:

| | |
|---|---|
| a. Temperature | 73° ± 18°F |
| b. Relative humidity | 50 percent ± 30 percent |
| c. Atmospheric pressure | 28.5 + 2.0 -3.0 in. Hg. |

3.1.4 Reference axis. Zero reference in azimuth, or forward position of the sensor, is defined as follows: Observe the electrical connector. The direction of pin 6 from the connector's centerline indicates the forward direction, zero degrees azimuth. Full crosswind then occurs at plus 90 and plus 270 degrees from the zero point (See figure 1).

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.2).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under the following conditions:

| Item | Condition |
|---|---|
| a. Standard ambient | see 3.1.3 |
| b. Input voltage | 15.000 ± 0.030 volts direct current (vdc)<br>-15.000 ± 0.030 vdc<br>28.000 ± 0.250 vdc |
| c. Electrical set-up | see figure 2 |
| d. Wind velocity tolerance | ± 10 percent of specified values |



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3.3.1 Zero wind operation. Under a zero wind environment and with the electronics card (EC) adjusted for a voltage between Bridge 3/Signal B (pin 2-CWS) and Bridge 2 (pin 11-CWS) of 0 ± 100 microvolts dc;

a. The voltage between Bridge 1 (pin 21-CWS) and power return (pin 23-CWS) shall be 3.16 ± 0.10 vdc.



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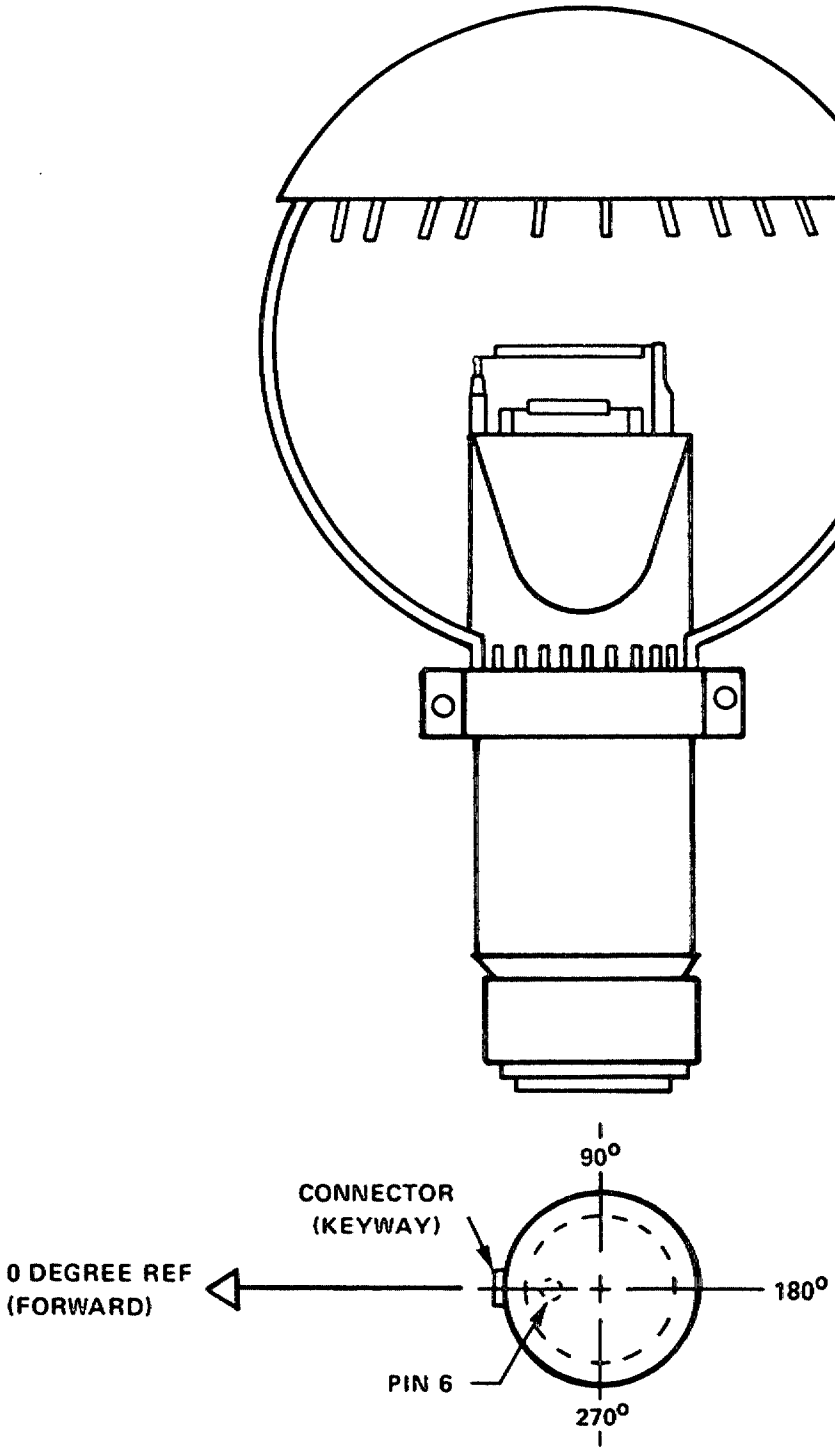



FIGURE 1. Zero degree reference orientation.



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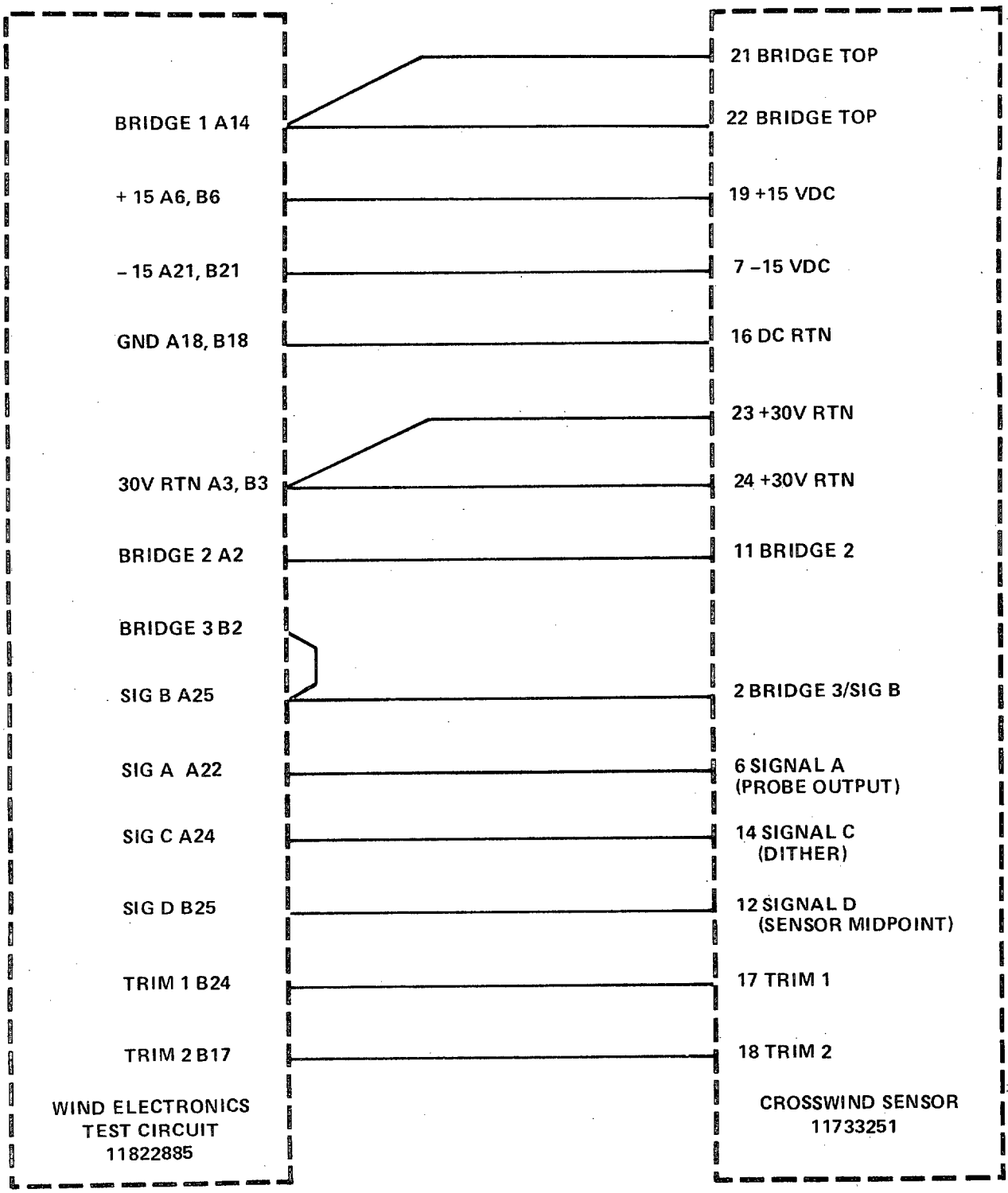



FIGURE 2. Electrical connections to crosswind sensor.



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b. The voltage between signal A (pin 6-CWS) and dc return (pin 16-CWS) shall be 0.0 ± 1 vdc.

c. The voltage between test point 10 on the electronics card (TP10-EC) and signal return (pin A18-EC) shall be 0.00 ± 0.05 vdc. (See 4.6.2.1)

3.3.2 Operational requirements. With specified air velocities at specified azimuth angles and at zero roll and pitch angles, the assembly shall provide output voltages between $V_w$ (pin A17-EC) and dc return (pin 16-CWS) within the limits specified in table I and table II. One hundred percent of the output voltages of table I shall be met and 68 percent of the output voltages of table II shall be met. (See 4.6.2.2)

3.3.2.1 Pitch. At a pitch angle of 15 degrees, either positive or negative, for wind velocities of 10 and 40 miles per hour (mph), the assembly shall provide output voltages within the limits specified in table I.

3.3.2.2 Roll. At a roll angle of 15 degrees, either positive or negative, for wind velocities of 10 and 40 mph, the assembly shall provide output voltages within the limits specified in table I.

3.3.3 Stability. With a constant wind velocity of 20 mph having aspect angles of 90 degrees and 270 degrees (full crosswind) with respect to the front of the assembly, the output voltage shall remain within 0.200 volts of its initial value for not less than 10 minutes. The dc voltage shall be measured between $V_w$ (pin A17-EC) and the return (pin A18-EC). (see 4.6.2.3)

3.4 Environmental.

3.4.1 Shock. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to three half sine wave shock pulses of 40 ± 4 gravity unit (g) for a duration of 18 ± 3 milliseconds (msec) applied in each direction along three mutually perpendicular axes. In addition, the assembly shall be capable of withstanding three half sine wave shock pulses of 125 ± 12.5 g for a duration of 5.0 ± 0.2 msec applied in each direction along three mutually perpendicular axes. (See 4.6.3.1)

3.4.2 Vibration. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to the vibration profile of figure 3. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes. (See 4.6.3.2)



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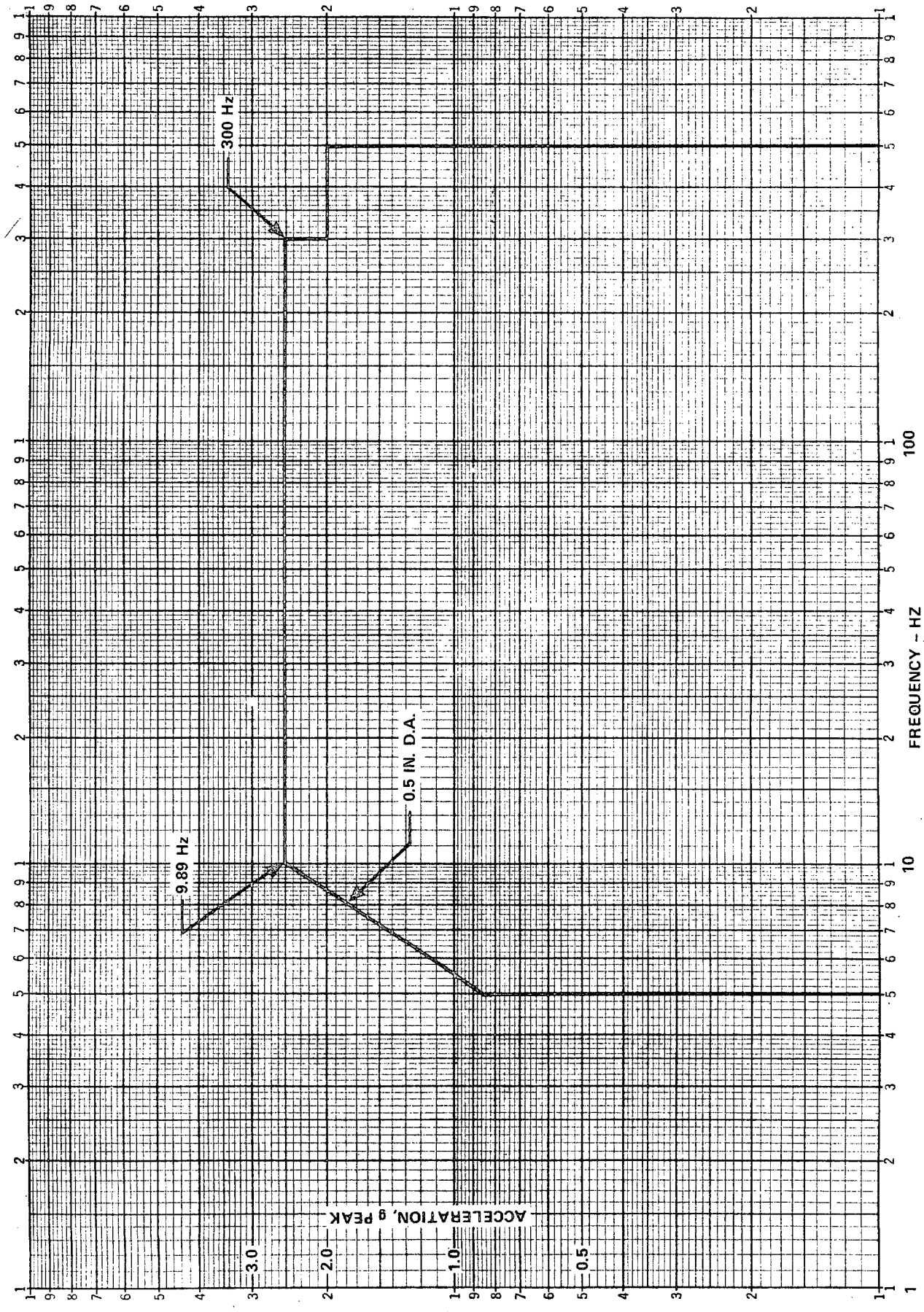



FIGURE 3. Vibration profile.



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TABLE I. Wind sensor performance (100%).

| Orientation of Wind Sensor to Wind in Degrees | Output Voltage (DC) At Wind Velocities, $W_c$ | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 5 MPH | | 10 MPH | | 20 MPH | | 30 MPH | | 40 MPH | |
| | Between | And | Between | And | Between | And | Between | And | Between | And |
| 270 | | | | | 2.800 | -5.200 | | | -7.90 | -8.10 |
| 275 | -0.566 | -1.366 | | | | | | | | |
| 280 | | | -1.368 | -2.568 | | | -4.104 | -7.704 | | |
| 285 | | | | | -2.660 | -5.060 | | | | |
| 290 | -0.539 | -1.339 | | | | | | | -5.11 | -9.91 |
| 295 | | | -1.212 | -2.412 | | | | | | |
| 300 | | | | | -2.264 | -4.664 | | | | |
| 305 | -0.419 | -1.219 | | | | | | | | |
| 310 | | | -0.938 | -2.138 | | | -2.796 | -6.396 | | |
| 315 | | | | | -1.628 | -4.028 | | | | |
| 320 | -0.242 | -1.042 | | | | | | | -2.73 | -7.53 |
| 325 | | | -0.546 | -1.746 | | | | | | |
| 330 | | | | | -0.800 | -3.200 | | | | |
| 335 | -0.022 | -0.822 | | | | | | | | |
| 340 | | | -0.084 | -1.284 | | | -0.252 | -3.852 | | |
| 345 | | | | | +0.168 | -2.232 | | | | |
| 350 | +0.227 | -0.573 | | | | | | | +1.02 | -3.78 |
| 355 | | | +0.426 | -0.774 | | | | | | |
| 360/0 | | | | | +1.200 | -1.200 | | | | |



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TABLE I. Wind sensor performance (100%). - Continued

| Orientation of Wind Sensor to Wind in Degrees | Output Voltage (DC) At Wind Velocities, $W_c$ | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 5 MPH | | 10 MPH | | 20 MPH | | 30 MPH | | 40 MPH | |
| | Between | And | Between | And | Between | And | Between | And | Between | And |
| 5 | +0.487 | -0.313 | | | | | | | | |
| 10 | | | +0.946 | -0.254 | | | +2.838 | -0.762 | | |
| 15 | | | | | +2.232 | +0.168 | | | | |
| 20 | +0.742 | -0.058 | | | | | | | +5.13 | +0.33 |
| 25 | | | +1.444 | +0.244 | | | | | | |
| 30 | | | | | +3.200 | +0.800 | | | | |
| 35 | +0.973 | +0.173 | | | | | | | | |
| 40 | | | +1.884 | +0.684 | | | +5.652 | +2.052 | | |
| 45 | | | | | +4.028 | +1.628 | | | | |
| 50 | +1.166 | +0.366 | | | | | | | +8.52 | +3.72 |
| 55 | | | +2.283 | +1.083 | | | | | | |
| 60 | | | | | +4.664 | +2.264 | | | | |
| 65 | +1.306 | +0.506 | | | | | | | | |
| 70 | | | +2.478 | +1.278 | | | +7.434 | +3.834 | | |
| 75 | | | | | +5.060 | +2.660 | | | +10.27 | +5.47 |
| 80 | +1.384 | +0.584 | | | | | | | | |
| 85 | | | +2.592 | +1.392 | | | | | | |
| 90 | | | | | +5.200 | +2.800 | | | +8.10 | +7.90 |

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TABLE I. Wind sensor performance (100%). - Continued

| Orientation of Wind Sensor to Wind in Degrees | Output Voltage (DC) At Wind Velocities, $W_c$ | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 5 MPH | | 10 MPH | | 20 MPH | | 30 MPH | | 40 MPH | |
| | Between | And | Between | And | Between | And | Between | And | Between | And |
| 95 | +1.366 | +0.566 | | | | | | | | |
| 100 | | | +2.568 | +1.368 | | | +7.704 | +4.104 | | |
| 105 | | | | | +5.060 | +2.660 | | | | |
| 110 | +1.339 | +0.539 | | | | | | | +9.91 | +5.11 |
| 115 | | | +2.412 | +1.212 | | | | | | |
| 120 | | | | | +4.664 | +2.264 | | | | |
| 125 | +1.219 | +0.419 | | | | | | | | |
| 130 | | | +2.138 | +0.938 | | | +6.396 | +2.796 | | |
| 135 | | | | | +4.028 | +1.628 | | | | |
| 140 | +1.042 | +0.242 | | | | | | | +7.53 | +2.73 |
| 145 | | | +1.746 | +0.546 | | | | | | |
| 150 | | | | | +3.200 | -0.860 | | | | |
| 155 | +0.822 | +0.022 | | | | | | | | |
| 160 | | | +1.284 | +0.084 | | | | | | |
| 165 | | | | | +2.232 | -0.168 | | | | |
| 170 | +0.573 | -0.227 | | | | | | | +3.78 | -1.02 |
| 175 | | | +0.774 | -0.426 | | | | | | |
| 180 | | | | | +1.200 | -1.200 | | | | |
| 185 | +0.313 | -0.487 | | | | | | | | |
| 190 | | | +0.254 | -0.946 | | | +0.762 | -2.838 | | |
| 195 | | | | | +0.168 | -2.232 | | | | |



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TABLE I. Wind sensor performance (100%) - Continued.

| Orientation of Wind Sensor to Wind in Degrees | Output Voltage (DC) At Wind Velocities, $W_c$ | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 5 MPH | | 10 MPH | | 20 MPH | | 30 MPH | | 40 MPH | |
| | Between | And | Between | And | Between | And | Between | And | Between | And |
| 200 | +0.058 | -0.742 | | | | | | | -0.33 | -5.13 |
| 205 | | | -0.244 | -1.444 | | | | | | |
| 210 | | | | | -0.800 | -3.200 | | | | |
| 215 | -0.173 | -0.973 | | | | | | | | |
| 220 | | | -0.684 | -1.884 | | | -2.052 | -5.652 | | |
| 225 | | | | | -1.628 | -4.028 | | | | |
| 230 | -0.366 | -1.166 | | | | | | | -3.72 | -8.52 |
| 235 | | | -1.083 | -2.283 | | | | | | |
| 240 | | | | | -2.264 | -4.664 | | | | |
| 245 | -0.506 | -1.306 | | | | | | | | |
| 250 | | | -1.278 | -2.478 | | | -3.834 | -7.434 | | |
| 255 | | | | | -2.660 | -5.060 | | | | |
| 260 | -0.584 | -1.384 | | | | | | | -5.47 | -10.27 |
| 265 | | | -1.392 | -2.592 | | | | | | |
| 270 | | | | | -2.800 | -5.200 | | | -7.90 | -8.10 |



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TABLE II. Wind sensor performance (68%).

| Orientation of Wind Sensor to Wind in Degrees | Output Voltage (DC) At Wind Velocities, $W_c$ | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 5 MPH | | 10 MPH | | 20 MPH | | 30 MPH | | 40 MPH | |
| | Between | And | Between | And | Between | And | Between | And | Between | And |
| 270 | | | | | -3.400 | -4.600 | | | -7.900 | -8.100 |
| 275 | -0.766 | -1.166 | | | | | | | | |
| 280 | | | -1.668 | -2.268 | | | -5.004 | -6.804 | | |
| 285 | | | | | -3.260 | -4.460 | | | | |
| 290 | -0.739 | -1.139 | | | | | | | -6.310 | -8.710 |
| 295 | | | -1.512 | -2.112 | | | | | | |
| 300 | | | | | -2.864 | -4.064 | | | | |
| 305 | -0.619 | -1.019 | | | | | | | | |
| 310 | | | -1.238 | -1.838 | | | -3.696 | -5.496 | | |
| 315 | | | | | -2.228 | -3.428 | | | | |
| 320 | -0.442 | -0.842 | | | | | | | -3.930 | -6.330 |
| 325 | | | -0.846 | -1.446 | | | | | | |
| 330 | | | | | -1.400 | -2.600 | | | | |
| 335 | -0.222 | -0.622 | | | | | | | | |
| 340 | | | -0.384 | -0.984 | | | -1.152 | -2.952 | | |
| 345 | | | | | -0.432 | -1.632 | | | | |
| 350 | +0.027 | -0.373 | | | | | | | -0.180 | -2.580 |
| 355 | | | +0.126 | -0.474 | | | | | | |
| 360 | | | | | +0.600 | -0.600 | | | | |
| 5 | +0.287 | -0.113 | | | | | | | | |
| 10 | | | +0.646 | +0.046 | | | +1.938 | +0.138 | | |



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TABLE II. Wind sensor performance (68%). - Continued

| Orientation of Wind Sensor to Wind in Degrees | Output Voltage (DC) At Wind Velocities, $W_c$ | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 5 MPH | | 10 MPH | | 20 MPH | | 30 MPH | | 40 MPH | |
| | Between | And | Between | And | Between | And | Between | And | Between | And |
| 15 | | | | | +1.632 | +0.432 | | | | |
| 20 | +0.542 | +0.142 | | | | | | | +3.930 | +1.530 |
| 25 | | | +1.144 | +0.544 | | | | | | |
| 30 | | | | | +2.600 | +1.400 | | | | |
| 35 | +0.773 | +0.373 | | | | | | | | |
| 40 | | | +1.584 | +0.984 | | | +4.752 | +2.952 | | |
| 45 | | | | | +3.428 | +2.223 | | | | |
| 50 | +0.966 | +0.566 | | | | | | | +7.320 | +4.920 |
| 55 | | | +1.983 | +1.383 | | | | | | |
| 60 | | | | | +4.064 | +2.864 | | | | |
| 65 | +1.106 | +0.706 | | | | | | | | |
| 70 | | | +2.178 | +1.578 | | | +6.534 | +4.734 | | |
| 75 | | | | | +4.460 | +3.260 | | | | |
| 80 | +1.184 | +0.784 | | | | | | | +9.070 | +6.770 |
| 85 | | | +2.292 | +1.692 | | | | | | |
| 90 | | | | | +4.600 | +3.400 | | | +8.100 | +7.900 |
| 95 | +1.166 | +9.766 | | | | | +6.804 | +5.004 | | |
| 100 | | | +2.268 | +1.668 | | | | | | |
| 105 | | | | | +4.460 | +3.260 | | | | |
| 110 | +1.139 | +0.739 | | | | | | | +8.710 | +6.310 |
| 115 | | | +2.112 | +1.512 | | | | | | |
| 120 | | | | | +4.064 | +2.864 | | | | |



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TABLE II. Wind sensor performance (68%). - Continued

| Orientation of Wind Sensor to Wind in Degrees | Output Voltage (DC) At Wind Velocities, $W_c$ | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 5 MPH | | 10 MPH | | 20 MPH | | 30 MPH | | 40 MPH | |
| | Between | And | Between | And | Between | And | Between | And | Between | And |
| 125 | +1.019 | +0.619 | | | | | | | | |
| 130 | | | +1.838 | +1.238 | | | +5.496 | +3.696 | | |
| 135 | | | | | +3.428 | +2.228 | | | | |
| 140 | +0.842 | +0.442 | | | | | | | +6.330 | +3.930 |
| 145 | | | +1.446 | +0.846 | | | | | | |
| 150 | | | | | +2.600 | +1.400 | | | | |
| 155 | +0.622 | +0.222 | | | | | | | | |
| 160 | | | +0.984 | +0.384 | | | +2.952 | +1.152 | | |
| 165 | | | | | +1.632 | +0.432 | | | | |
| 170 | +0.373 | -0.027 | | | | | | | | |
| 175 | | | +0.474 | -0.126 | | | | | | |
| 180 | | | | | +0.600 | -0.600 | | | | |
| 185 | +0.113 | -0.287 | | | | | | | | |
| 190 | | | -0.046 | -0.646 | | | -0.138 | -1.938 | | |
| 195 | | | | | -0.432 | -1.632 | | | | |
| 200 | -0.142 | -0.542 | | | | | | | -1.530 | -3.930 |
| 205 | | | -0.544 | -1.144 | | | | | | |
| 210 | | | | | -1.400 | -2.600 | | | | |
| 215 | -0.373 | -0.773 | | | | | | | | |
| 220 | | | -0.984 | -1.584 | | | -2.952 | -4.752 | | |



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TABLE II. Wind sensor performance (68%). - Continued

| Orientation of Wind Sensor to Wind in Degrees | Output Voltage (DC) At Wind Velocities, $W_c$ | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 5 MPH | | 10 MPH | | 20 MPH | | 30 MPH | | 40 MPH | |
| | Between | And | Between | And | Between | And | Between | And | Between | And |
| 225 | | | | | -2.228 | -3.428 | | | | |
| 230 | -0.566 | -0.966 | | | | | | | -4.920 | -7.320 |
| 235 | | | -1.383 | -1.983 | | | | | | |
| 240 | | | | | -2.864 | -4.064 | | | | |
| 245 | -0.706 | -1.106 | | | | | | | | |
| 250 | | | -1.578 | -2.178 | | | -4.734 | -6.534 | | |
| 255 | | | | | -3.260 | -4.460 | | | | |
| 260 | -0.784 | -1.084 | | | | | -6.670 | -9.070 | -6.670 | -9.070 |
| 265 | | | -1.692 | -2.292 | | | | | | |
| 270 | | | | | -3.400 | -4.600 | | | -7.900 | -8.100 |



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3.4.3 Operating temperature. The assembly shall be capable of operating as specified herein over the operating temperature range of -40°F to +125°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

3.4.4 Storage temperature. The assembly shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65°F to 160°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5).

3.4.5 Humidity. The assembly shall operate as specified herein at up to 100 percent relative humidity. (See 4.6.3.6)

3.4.6 Altitude.

3.4.6.1 Dynamic. The assembly shall operate as specified herein during exposure to an atmospheric pressure equivalent to 10000 feet above sea level.

3.4.6.2 Static. The assembly shall operate as specified herein after exposure to an atmostpheric pressure equivalent to 40000 feet above sea level for not less than 1 hour. (See 4.6.3.7)

3.4.7 Dust. The assembly shall operate as specified herein after exposure to moving dust for not less than 12 hours at velocities up to 1750 feet per minute. (See 4.6.3.8)

3.4.8 Salt fog. The assembly shall operate as specified herein after exposure to a 5 percent sodium chloride atomized spray for not less than 48 hours. (See 4.6.3.9)

3.4.9 Waterproofness. The assembly shall be capable of withstanding not less than 8.7 pounds per square inch (psi) differential pressure while submerged in water for not less than 2 hours. (See 4.6.3.10)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth herein where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.



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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (See 4.4).
2. Quality conformance inspection (See 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

4.4 First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three assemblies selected at random by the Government representative from the first fifteen production assemblies.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in table III.

4.4.3 Failure. Failure of any assembly to meet any requirement shall be cuase for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any assembly to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.



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Table III. First article tests.

| Item | Characteristics | Requirement | Test Procedure |
|---|---|---|---|
| 1 | Fabrication | 3.1 | Applicable drawings-Visual |
| 2 | General specification | 3.1.2 | MIL-F-13926 Visual |
| 3 | Temperature cycling | 3.4.3 and 3.4.4 | 4.6.3.5 |
| 4 | Zero wind operation | 3.3.1 | 4.6.2.1 |
| 5 | Operational requirements | 3.3.2 | 4.6.2.2 |
| 6 | Stability | 3.3.3 | 4.6.2.3 |
| 7 | Shock | 3.4.1 | 4.6.3.1 |
| 8 | Vibration A | 3.4.2 | 4.6.3.2.1 |
| 9 | High temperature | 3.4.3 and 3.4.4 | 4.6.3.3 |
| 10 | Low temperature | 3.4.3 and 3.4.4 | 4.6.3.4 |
| 11 | Humidity | 3.4.5 | 4.6.3.6 |
| 12 | Altitude | 3.4.6 | 4.6.3.7 |
| 13 | Dust | 3.4.7 | 4.6.3.8 |
| 14 | Salt fog | 3.4.8 | 4.6.3.9 |
| 15 | Waterproofness | 3.4.9 | 4.6.3.10 |



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4.5 Quality conformance inspection.

4.5.1 Procurement conditions. The following inspection plans shall apply where the assembly is procured for use as a:

| Use | Plan |
|---|---|
| a. Logistics spare assembly apart from the next higher assembly | A and C |
| b. Component assembly of the next higher assembly | B and C |

4.5.2 Inspection plan A.

4.5.2.1 General sample and tests. One assembly, as a control sample, shall be selected at random by the Government representative from each 100 assemblies produced and shall be subjected to all the tests in table V.

4.5.2.2 Environmental sample and tests. Three assemblies, as a control sample, shall be selected at random by the Government representative from each 50 assemblies produced or from each month's production, whichever occurs first. All the tests in table VI shall be applied separately to each assembly in the sample.

4.5.2.3 Acceptance. Where any one assembly of either sample fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subject to the provisions of MIL-STD-105.

4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative in accordance with the provisions of MIL-STD-105. All the tests in table V and only the test of item 307 in table VI shall be applied.

4.5.3.2 Acceptance. Where any one assembly fails to meet any specified requirement, the defective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each assembly in every lot shall be subjected to all the tests in table IV and shall be examined visually for completeness, improper assembly and evidence of poor workmanship.

4.5.4.2 Acceptance. Where any one assembly fails to meet any specified requirement, the defective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.



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4.5.5 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662.

4.5.6 Packaging inspection. The sampling and inspection of the preservation - packaging, packing and container marking shall be in accordance with the provisions of packaging data sheet P11733251.

TABLE IV. Performance tests.

| Item | Characteristic | Requirement | Test Procedure |
|---|---|---|---|
| 101 | Zero wind operation | 3.3.1 | 4.6.2.1 |
| 102 | Operational requirements | 3.3.2 | 4.6.2.2 |
| 103 | Stability | 3.3.3 | 4.6.2.3 |

TABLE V. General tests.

| Item | Characteristic | Requirement | Test Procedure |
|---|---|---|---|
| 301 | Fabrication | 3.1 | Applicable drawings-Visual |
| 302 | General specification | 3.1.2 | MIL-F-13926 Visual |



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TABLE VI. Environmental tests.

| Item | Characteristic | Requirement | Test Procedure |
|---|---|---|---|
| 303 | Vibration B | 3.4.2 | 4.6.3.2.2 |
| 304 | High temperature | 3.4.3 and 3.4.4 | 4.6.3.3 |
| 305 | Low temperature | 3.4.3 and 3.4.4 | 4.6.3.4 |
| 306 | Waterproofness | 3.4.9 | 4.6.3.10 |
| 307 | Temperature cycling | 3.4.3 and 3.4.4 | 4.6.3.5 |
| 308 | Shock | 3.4.1 | 4.6.3.1 |

4.6 Methods of inspection.

4.6.1 Test equipment. The test equipment in table VII shall be used where specified to perform the required test.

TABLE VII. Test equipment.

| Item 1/ | Part number | Nomenclature |
|---|---|---|
| 1 | 11822864 | Test Set - Crosswind Sensor |
| 2 | 11822890 | Shock and Vibration Fixture Assembly |
| 3 | 11822889 | Shock Fixture, Sensor Mast Mounting Plate |
| 4 | 11822891 | Cable, Wiring Diagram |
| 5 | 11733250 | Mast, Crosswind Sensor |
| 6 | 11821463 | Unit Fixture - Crosswind Sensor |
| 7 | 11821464 | Unit Fixture - Crosswind Sensor (-Z) |

1/ Items 2, 3, 4, 5, 6, 7 are parts of 11750233, Test Set - Computer System FAT

4.6.2 Performance tests.

4.6.2.1 Zero wind operation. Use item 1 of table VII to verify compliance with the requirements of 3.3.1.

4.6.2.2 Operational requirements. Use item 1 of table VII to verify compliance with the requirements of 3.3.2.



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4.6.2.3 Stability. Use item 1 of table VII to verify compliance with the requirements of 3.3.3.

4.6.3 Environmental tests.

4.6.3.1 Shock.

4.6.3.1.1 Basic design. Use items 2 and 3 of table VII and subject the assembly to the shock test specified in MIL-STD-810, method 516.2, procedure I, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 40 ±4 g with a time duration of 18 ±3 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table IV.

4.6.3.1.2 High intensity. Use items 2 and 3 of table VII and subject the assembly to the shock test specified in MIL-STD-810, method 516.2, procedure IV, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 125.0 ± 12.5 g with a time duration of 5.0 ±0.2 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table IV.

4.6.3.2 Vibration. Application of Vibration A and Vibration B shall be as determined by the procurement conditions of 4.5.1 and the first article conditions of 4.4.

4.6.3.2.1 Vibration A. Use item 2 of table VII and subject the assembly to the vibration test specified in MIL-STD-810, method 514.2 procedure VIII, except that the test level shall be the vibration curve shown in figure 3 and duration of exposure shall be not less than 80 minutes per axis. At the conclusion of this test, subject the assembly to the performance tests of table IV.

4.6.3.2.2 Vibration B. This test shall be conducted as in 4.6.3.2.1 except that the period of vibration shall be not less than 15 minutes in each axis.

4.6.3.3 High temperature. Subject the assembly to the high temperature test specified in MIL-STD-810, method 501.1, procedure I. The highest operating temperature shall be 125°F. The performance tests of table IV shall be applied.

4.6.3.4 Low temperature. Subject the assembly to the low temperature test specified in MIL-STD-810, method 502.1, procedure I. The storage temperature shall be -65°F. The lowest operating temperature shall be -40°F. The performance tests of table IV shall be applied.



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4.6.3.5 Temperature cycling. With the input voltages and signals of 3.3 applied subject the assembly to the temperature profile specified in figure 4. At the conclusion of this test, subject the assembly to the performance tests of table IV.

4.6.3.6 Humidity. Use item 1 of table VII, place the assembly in a humidity chamber and subject the assembly to the humidity test specified in MIL-STD-810, method 507.1, procedure IV. During steps 4, 6 and 8 of procedure IV, subject the assembly to the performance tests of table IV.

4.6.3.7 Altitude. Use item 1 of table VII, place the assembly in an altitude chamber and subject it to the altitude test per MIL-STD-810, method 500.1, procedure 1, except that:

a. First, the assembly in a non-operating condition, shall be exposed to an atmospheric pressure equivalent to 40000 feet above sea level for a period of not less than 1 hour.

b. Second, with the chamber pressure increased to an atmospheric equivalent to 10000 feet above sea level, subject the assembly to the performance tests of table IV.

4.6.3.8 Dust. Use item 1 of table VII and subject the assembly to the dust test specified in MIL-STD-810, method 510.1, procedure I. At the conclusion of this exposure, subject the assembly to the performance tests of table IV.

4.6.3.9 Salt fog. Use item 1 of table VII and subject the assembly to the corrosion test specified in MIL-STD-810, method 509.1, procedure I. At the conclusion of this exposure, subject the assembly to the performance tests of table IV.

4.6.3.10 Waterproofness. Submerge the assembly in room temperature water within a pressure chamber. Raise the pressure in the chamber to not less than 8.7 psi and maintain that pressure for not less than two hours. After exposure, disassemble the assembly and inspect for water inside.



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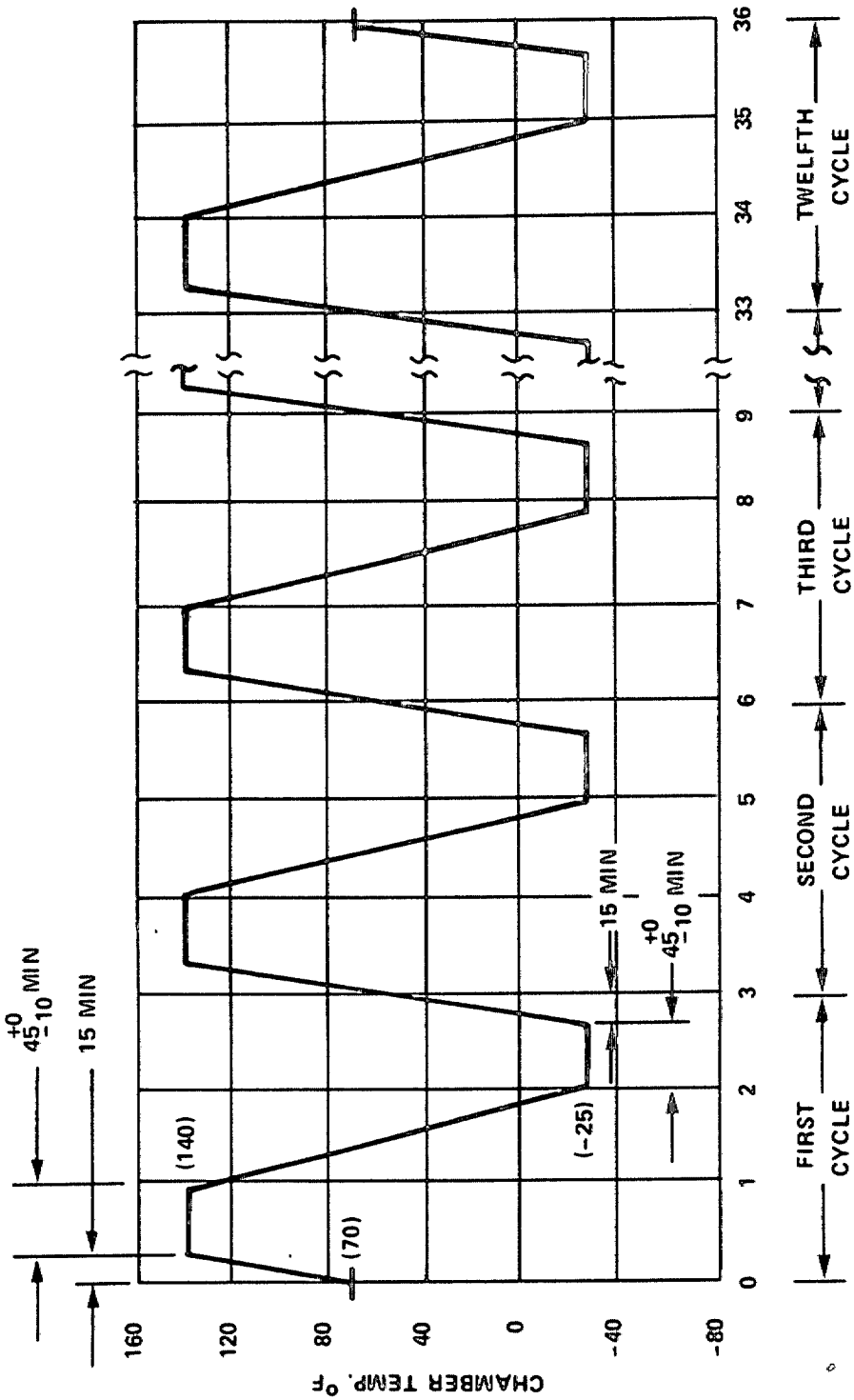



FIGURE 4. Temperature cycling.



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## 5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with packaging data sheet P11733251. The level of protection shall be as specified in the procurement document. (See 4.5.6)

## 6. NOTES

6.1 Intended use. The crosswind sensor is a part of the M21 Ballistic Computer System.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Quantity required and delivery schedules.

c. Special tests and examination as required, the type of inspection required and the activity responsible for testing and its facility.

d. Serialization requirements, if applicable.

e. Requirements for first article sample, including data requirements for certificates of conformance for materials.

f. Applicable packaging data sheet number.

g. Selection of applicable levels of preservation, packaging and packing.

h. Applicable stock number.

6.3 Previous revisions of MIL-S-48464B was for the US Army Armament Research and Development Command's internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

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